CIVORA

AI-Powered Grievance Redressal & Management Platform

Every grievance deserves a resolution.

THE CHALLENGE

A civic problem is often reported by someone standing at the site, using a phone and speaking the language they are most comfortable with. Long text forms can make reporting harder, while complaints may still need manual sorting, departmental review and repeated follow-up. Citizens can struggle to understand what happens after filing, and administrators need a consolidated view of work across departments and locations.

THE CIVORA SOLUTION

CIVORA provides a single digital workflow for civic grievances, from citizen intake to staff review and closure. Its defining approach is **voice-first complaint drafting**: citizens describe an issue in their own words, speech is converted into a structured written draft, and the citizen can review, edit or retake the transcript before submitting it. A typed form remains available. Submitted grievances enter an auditable, role-based process for review, routing, assignment, updates and resolution.

VOICE EXPERIENCE POWERED BY GNANI

Gnani Prisma v3.0 provides speech-to-text transcription for guided, turn-by-turn answers and returns a transcript with detected language. **Gnani Timbre** provides spoken intake prompts and optional spoken responses. Audio is not submitted automatically: the citizen reviews and approves the written details first. The current citizen interface offers English, Hindi and Marathi; Prisma supports ten Indian languages, creating a path for broader language coverage.

CORE CAPABILITIES

Accessible citizen intake	**AI-assisted triage**	**Role-based grievance workflow**
Voice or typed filing; optional incident location, map selection and image evidence; unique reference IDs; status timeline and in-app notifications.	Local scikit-learn classification and keyword rules support categorization and triage. Optional Hugging Face and Groq fallbacks can be configured. AI recommendations are advisory.	Citizens track cases; global administrators review and route; department managers allocate work; employees update assigned tasks and submit resolutions for manager review.
Department visibility	**Maps and analytics**	**Account and service options**
Global and department-scoped workspaces provide grievance queues, employee allocation and department-specific analytics.	Interactive charts and location maps help staff examine complaint distribution and understand where grievances are reported.	JWT sessions and server-side role checks; optional phone OTP, facial sign-in, Cloudinary image uploads and Twilio SMS notifications, subject to configuration.

END-TO-END WORKFLOW

1. Report: A signed-in citizen describes the issue by voice or text and adds available location details or evidence.
2. Review: The citizen checks the written grievance.
3. Triage: Backend classification and rules assist categorization.
4. Act: Authorized staff review routing, managers assign employees, and employees post updates.
5. Resolve: Employees submit completed work for manager review before the grievance proceeds to closure.

TECHNOLOGY AND DEPLOYMENT

Frontend: Next.js 16, React 19, TypeScript, Tailwind CSS 4 | **Backend:** Python, FastAPI, Pydantic, Uvicorn
Data: MongoDB via PyMongo | **AI/text:** scikit-learn and keyword rules; optional Hugging Face and Groq
Speech: Gnani Prisma v3.0 STT and Gnani Timbre TTS | **Visualization:** Leaflet and Recharts
Live prototype: civora-public-grievance.vercel.app | **Hosting:** Vercel frontend, Render API and MongoDB Atlas

VALUE, RESPONSIBLE AI AND NEXT STEPS

CIVORA aims to make complaint filing more accessible, improve visibility into case progress, reduce manual sorting effort and help departments coordinate responses. **Human oversight is central:** AI does not make final decisions about routing, employee assignment, escalation, resolution approval or closure; authorized people do. As a working prototype, CIVORA still requires broader testing for speech quality in noisy environments, classification accuracy, high concurrent usage and real-world administrative deployment. Future work includes expanded language evaluation, load testing, predictive grievance insights and integration with external government systems.